

Bearing remaining useful life prediction with an improved CNN-LSTM network using an artificial gorilla troop optimization algorithm

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Abstract

To address the problem of reliance on a priori knowledge and difficult hyperparameter selection in feature fusion. The effect of different convolutional kernel sizes and filters on feature fusion is investigated firstly, based on which an artificial Gorilla Troops Optimizer (GTO) enhanced Convolutional Long-Short Term Memory Neural Network (CNN-LSTM) method for bearing lifetime prediction is suggested. The GTO algorithm was used to optimize hyperparameters such as the convolutional kernel size of CNN-LSTM, filters and pooling layer size, batch size, number of hidden layer neurons, and rate of learning with the goal of minimizing the mean squared error of the remaining useful life (RUL) prediction. From the optimized CNN-LSTM network analyze the monitored performance degradation data, construct health indicators (HI) reflecting bearing degradation, and build the remaining bearing life prediction model. Typical life cycle data has been used for the validation of the proposed method. The results indicate that the health indicators have better trending and robustness, and leading to smaller errors in life prediction outcomes.

Keywords

Remaining useful life, gorilla troops optimizer, convolutional neural network, long short-term memory, health indicator

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Introduction

Rolling bearings are an important part of rotating equipment and can ensure the efficiency, precision, stability, and reliability of the mechanical transmission. Over time, under the influence of a combination of both external and internal factors, the behavior of bearings will deteriorate and eventually lead to equipment failure.¹ Relevant statistics indicate that nearly 30% of rotary machinery failures are caused by bearing failures.^{2,3} To ensure the safety and reliability of equipment, prognostics, and health management (PHM) of bearings can help to improve their safety and prevent catastrophic accidents.⁴ The remaining life prediction methods can be divided into three main groups: failure mechanism-based methods, monitoring data-based methods, and methods that integrate the two.^{5,6} The failure mechanism-based approach requires the establishment of a corresponding degradation model based on the specific object, which is generally not universal, and the increasing sophistication and complex equipment makes building an exact performance degradation

scheme too costly and limits the scope of application.⁷ The industrial Internet of Things (IoT) is quickly evolving to allow the collection of large amount of surveillance data through sensors. In recent years, deep learning has become one of the most active research topics, and Wu et al.⁸ proposed a diverse generation model to improve the efficiency and accuracy of foreign object detection in response to the problem of scarcity of image data for foreign object intrusion into transmission lines, thus affecting the detection accuracy. Xing et al.⁹ investigated the point cloud segmentation method in coal mining face (CMF) using deep learning

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techniques, and proposed the multi-level-dynamic graph convolution neural network (ML-DGCNN) model not only solved the defect of fixed neighborhood structure, but also enriched the local features of CMF point cloud, and finally verified the effectiveness of the proposed method by using coal cutting roof lines as an example. Zhao et al.¹⁰ propose a Dual Attention model (DATP) for predicting the near-term trajectories of road agents. Experiments demonstrate that the new hybrid CNN-LSTM network model simulating interactions between different road agents and introducing an attention mechanism to implicitly quantify the dynamic effects of the interactions on the future trajectories of the road agents can greatly improve the accuracy of trajectory prediction.

With the explosive growth in surveillance data, RUL prediction is facing new opportunities and challenges. Data-driven RUL prediction methods based on modeling degradation characteristics based on large amounts of historical bearing data typically consist of three stages, data collection, construction of health indicators, and life predictions.¹¹ A health indicator (HI) extracts valid characteristics from collected data and constructs them into indicators that can quantitatively depict the process of equipment degradation. Thus, RUL prediction accuracy is directly influenced by the construction of the health indicator.¹² To quantify the degradation process of equipment performance more accurately, many scholars have worked on health indicator construction methods. The main health indicators commonly used are root mean square, kurtosis,¹³ binary multiscale entropy,¹⁴ and Renyi entropy.¹⁵ In addition, the construction of bearing health indicators is gradually moving away from a single valid indicator to a fusion of multiple indicators extracted from the original signal. Hong et al.¹⁶ used WP-EMD to extract features from vibration signals and used self-organizing mapping (SOM) networks to fuse them to generate health indicators. Hu et al.¹⁷ used a deep belief network (DBN) to extract features from vibration signals, and local linear embedding (LLE) and Laplace feature mapping to fuse 29 time-frequency features into 1D features to construct health indicators.

Convolutional neural network (CNN) is widely used in the area of residual lifetime prediction, with features such as weight sharing, convolutional operations, and spatial pooling. Wang et al.¹⁸ used frequency domain features as the input feature set and combined CNN and long short-term memory neural network (LSTM) to predict the RUL of bearings. The proposed method constructed trend quantitative health indicators with good monotonic trends in both failure modes, but the CNN hyperparameter settings were not given in the paper. Mo et al.¹⁹ used a CNN network to construct three parallel convolutional channels to fuse features, and the resulting fused features contained richer information on bearing degradation, where the network hyperparameters set the number of filters to 32 and the convolutional kernel size to 6×1 . Jiang et al.²⁰ used

CNN, with network hyperparameters set to 32 filters and a convolutional kernel size of 100×1 , to extract feature sets sensitive to bearing degradation to improve prediction accuracy of bearing life. Zeng et al.²¹ used CNN networks to extract bearing features with temporal order and used training and fine-tuning layer by layer to improve parameter convergence. To enhance the prediction accuracy of bearing life for small samples, Zhang et al.⁷ proposed a bearing lifetime prediction method based on a full convolutional layer neural network. The aforementioned convolutional neural networks achieved excellent performance in the extraction of bearing features, but there were no detailed explanations for the setting of the network hyperparameters (convolutional kernel size, filter, pooling layer size, etc.), there is subjective and empirical dependence, and few scholars discussed the effects of different hyperparameters on feature extraction. Against this background, this paper discusses, for the first time, the effect of different hyperparameters on the feature extraction of CNN networks, based on which the GTO-CNN-LSTM network is proposed for RUL prediction. A health indicator describing the condition of the equipment is constructed by using a CNN-LSTM network optimized with the GTO algorithm that has adaptively selected hyperparameters to minimize the mean squared deviation of the lifetime prediction. Then, the remaining lifetime prediction model is constructed by using the optimized LSTM network to learn the degradation information of the health indicator to complete the lifetime prediction of the bearing.

The paper is structured as follows. In Section "Theoretical background," the basic theories of CNN, LSTM, and GTO are introduced separately. The proposed model to predict the remaining service life of rolling bearings is shown in Section "The proposed GTO-CNN-LSTM model for predicting the remaining useful life." In Section "Experimental demonstration," the effect of different hyperparameters on the features extracted by the CNN is explored, followed by validation of the accuracy of the proposed model using publicly available datasets. Conclusions are presented in Section "Conclusion."

Theoretical background

Convolutional neural network

A convolutional neural network is a network that uses convolutional operations. The convolutional layer can obtain characteristics from the initial data, the pooling layer selects and filters the characteristics obtained by the convolutional layer according to certain rules, and the fully connected layer is usually combined with the output layer to obtain the final dimensional change of the output. Models can extract spatial features from multiple samples, effectively capturing performance degradation information.²² The computation of the convolution layer proceeds as follows:

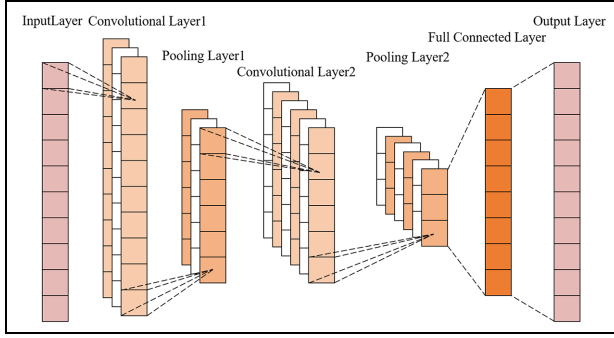


Figure 1. Structure of CNN network.

$$x_{i,j}^{l+1} = f\left(\sum_{i=1}^K \sum_{j=1}^n x_i^l \otimes W_j^i + b_i\right) \quad (1)$$

where $\otimes$ is the convolution operation, W_j^i is the i -th weight on the j -th convolution kernel, n is the convolution sum operation, f is the activation function, and $x_{i,j}^{l+1}$ is the i -th feature of the l -th layer x_i^l obtained by activating the i -th feature of the $l+1$ -th layer after the convolution kernel W_j^i operation with the activation function.

Figure 1 illustrates the structure of a typical convolutional neural network, which mainly consists of a convolutional layer, a pooling layer, and a fully connected layer.²³

Long short-term memory

Long and short-term memory networks are a variant of recurrent neural networks that can efficiently solve the gradient explosion or vanishing problem for simple recurrent neural networks.^{24–26} It takes into account the time relation between input and output and is widely used in rolling bearing condition prediction management. The LSTM consists of three gates, namely forgetting gate f_t , input gate i_t , and output gate o_t . These gate structures allow information to be selectively passed through, removed, or added to update the network. The memory cell c_t and the hidden state h_t in the LSTM are obtained by updating the previous states c_{t-1} and h_{t-1} with the following equations:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (2)$$

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (3)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (4)$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (5)$$

$$c_t = f_t * c_{t-1} + \tilde{c}_t * i_t \quad (6)$$

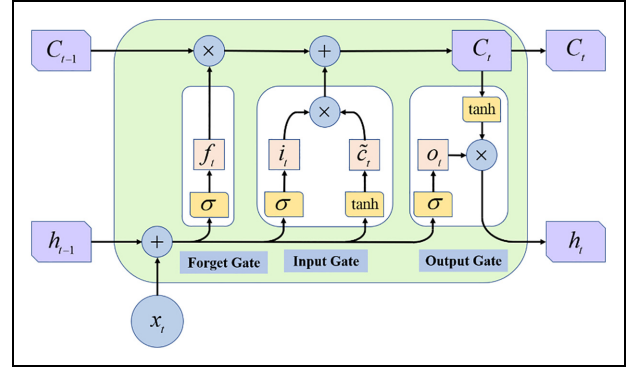


Figure 2. LSTM network structure.

$$h_t = o_t * \tanh(c_t) \quad (7)$$

where h_t is the output of the LSTM updating the network at time t , x_t represents the input signal, c_t is the internal state of the memory cell, c and h represent the memory cell and external state respectively, σ is the sigmoid function, $\tanh$ is the tanh function, $*$ represents the Hadamard product operator, W represents the weight of each neuron, and b is the bias vector of each gate cell.

Figure 2 shows the circular cell structure of the LSTM network.

Gorilla troops optimizer

GTO is a new intelligent optimization algorithm, mainly by studying the living performance of the whole gorilla to find the best. It has the characteristics of high discovery ability and high convergence efficiency.²⁷ Silverback gorillas lead a group, are responsible for the safety and welfare of the group, and determine the actions of the group, all gorillas obey the decisions of the silverback gorilla. As the silverback gorillas become weaker and older, the chimpanzee in the group may take over as group commander, or other male gorillas may attack the silverback gorillas and dominate the group. Three different mechanisms are used in the exploration phase, namely migration to unknown locations, migration to known sites, and migration to other gorillas. The mathematical model for the exploration phase is shown below:

$$GX(t+1) = \begin{cases} (UB - LB) \cdot r_1 + LB & rand < p \\ (r_2 - C) \cdot X_k(t) + L \cdot H, & rand \geq 0.5 \\ X(i) - L \cdot (L \cdot (X(t) - GX_k(t)) + r_3 \cdot (X(t) - GX_k(t))), & rand < 0.5 \end{cases} \quad (8)$$

$$C = F \cdot \left(1 - \frac{It}{MaxIt}\right) \quad (9)$$

$$F = \cos(2 \cdot r_4) + 1 \quad (10)$$

$$L = C \cdot l \quad (11)$$

$$H = Z \cdot X(t) \quad (12)$$

$$Z = [-C, C] \quad (13)$$

Where $GX(t+1)$ is the candidate position vector of the gorilla individual at the next iteration, $X(t)$ is the current position vector of the gorilla individual, r_1, r_2, r_3, r_4 and $rand$ are random numbers updated in the range 0–1 at each iteration. p is a given parameter between 0 and 1, which determines the probability of moving to an unknown location. UB, LB denote the upper and lower bounds of the variables, respectively. X_k, GX_k is a randomly selected groups of gorillas from the entire population. It is the current number of iterations, $MaxIt$ the maximum number of iterations, l is a random value between -1 and 1 , and Z is a random value in the problem dimension ranging from $[-C, C]$. The best solution generated at this stage is considered to be the silverback gorilla. The parameter W is set during the development phase, and if $C \geq W$, then it is chosen to follow the silverback gorilla, and equation (14) simulates this behavior:

$$GX(t+1) = L \cdot M \cdot (X(t) - X_{silverback}) + X(t) \quad (14)$$

$$M = \left(\left| \frac{1}{N} \sum_{i=1}^N GX_i(t) \right|^g \right)^{\frac{1}{g}} \quad (15)$$

where $X_{silverback}$ is the silverback gorilla position vector. $GX_i(t)$ refers to the position vector of each candidate gorilla at the t -th iteration, and N is the size of the gorilla population. If $C < W$, then select the second mechanism to develop:

$$GX(i) = X_{silverback} - (X_{silverback} \cdot (2 \cdot r_5 - 1) - X(t) \cdot (2 \cdot r_5 - 1)) \cdot \beta \cdot E \quad (16)$$

$$E = \begin{cases} N_1, & rand \geq 0.5 \\ N_2, & rand < 0.5 \end{cases} \quad (17)$$

where r_5 is a random value between 0 and 1, β is a given value of the parameter, if $rand < 0.5$, then E is a random value according to a normal distribution, otherwise E equals a random value according to a normal distribution and problem dimension.

The proposed GTO-CNN-LSTM model for predicting the remaining useful life

Since the construction of the Conventional CNN-LSTM network relies on empirical knowledge to select the hyperparameters, the GTO-CNN-LSTM network is used to select the optimal hyperparameters, performing adaptive feature fusion extracted from bearing vibration signals, mining deep bearing deterioration features to construct health indicators, and performing RUL prediction of bearings based on health indicators. The GTO-CNN-LSTM network is divided into a GTO part, a CNN-LSTM part, and a data part. The CNN-LSTM part decoded the incoming parameters of the GTO algorithm to obtain the required hyperparameters, then used the incoming training set of the data segment for network training. Finally, the prediction validation set obtained the mean squared error between the actual and expected output values and returns the mean squared error as the fitness value of the GTO algorithm. The algorithm optimizes according to the fitness value to achieve an update of the population with the global optimal solution. This method eventually yields the optimized network hyperparameters.

The RUL prediction scheme is shown in Figure 3. The horizontal vibration signal is input into the optimized CNN network after feature extraction to establish the health indicators, and then one part of the health indicators is input into the optimized LSTM network for training, and the other part is used as a test to finally get the RUL prediction results.

Step 1: Preprocessing of the data. The time domain, frequency domain, and time-frequency domain characteristics of the bearing vibration signal are extracted. Because of the large order of magnitude difference in the time and frequency domain characteristics, which results in unbalanced feature weights during merging, the original features are normalized using the min-max normalization method, with the following expressions:

$$x'_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (18)$$

where x is the original feature sample, x' is the normalized feature sample, $x_{\min}$ is the minimum feature sample value, and $x_{\max}$ is the maximum feature sample value.

Step 2: Construct bearing health indicators. Construct the CNN model, input the feature set obtained in Step 1 into the CNN structure for feature extraction, and obtain real-time features and failure features of the bearing. Construct the health indicators to measure the degree of bearing near failure state as follows²⁸:

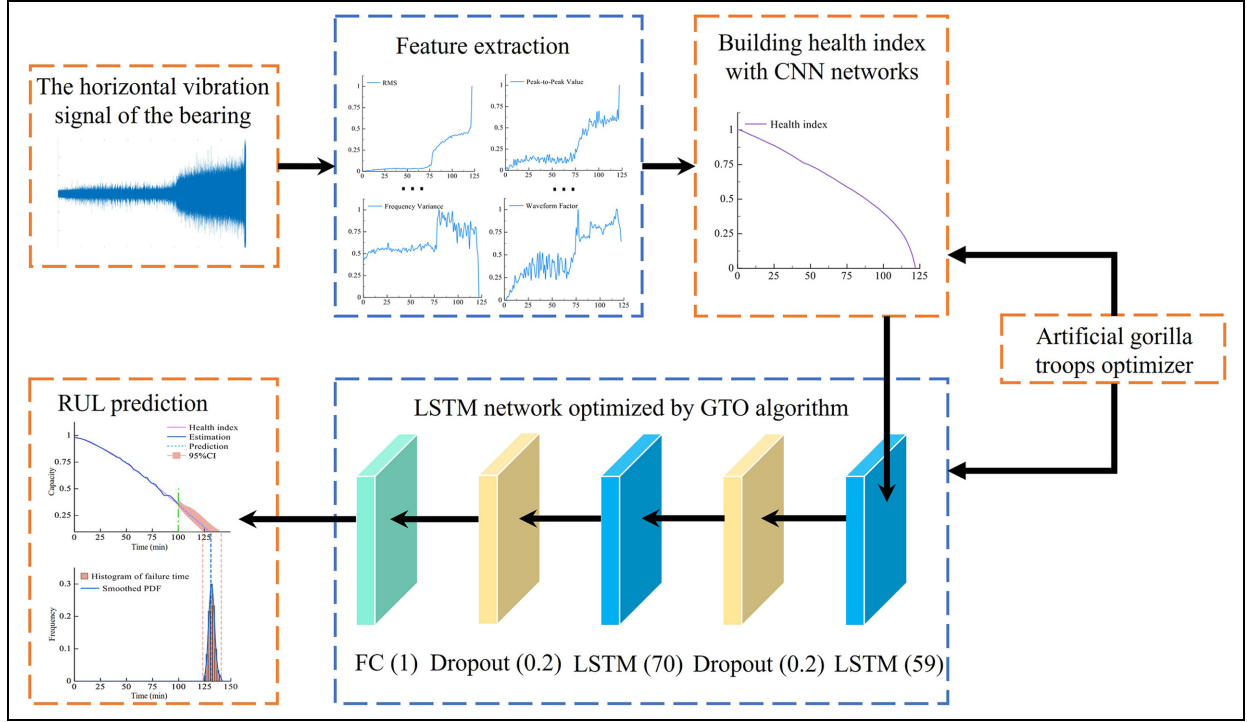


Figure 3. The framework of the proposed algorithm.

$$HI_k = \sqrt{\sum_{i=1}^K (f_i - f_{fault})^2} \quad (19)$$

where f_i is the real-time data feature, f_{fault} is the fault data feature, and K is the HI sequence length. The robustness and trend metrics used in the literature²⁹ are used to measure the HI to effectively evaluate the HI constructed by the proposed method and other methods.

(1) Robustness

HI curves typically contain random fluctuations due to measurement inaccuracies, the random nature of the

$$Rob(X) = \frac{1}{K} \sum_{n=1}^K \exp\left(-\left|\frac{x_n - x_n^T}{x_n}\right|\right) \quad (20)$$

where X is the HI series, K is the HI series length, x_n^T is the trend part of HI, and the trend part x_n^T and the residual part $x_n - x_n^T$ of HI are obtained using the sliding average method.

(2) Trend factors

Trend factors from the literature were used to describe the trend of the selected features quantitatively.^{31,32} The trend coefficient is an essential indicator of the connection between the selected features and time. The trend level of the extracted features can be determined by calculating the trend factor for each extracted feature.

$$Corr(X, T) = \frac{\left| K \sum_{n=1}^K x_n^T t_n - \sum_{n=1}^K x_n^T \sum_{n=1}^K t_n \right|}{\sqrt{\left[K \sum_{n=1}^K (x_n^T)^2 - \left(\sum_{n=1}^K x_n^T \right)^2 \right] \left[K \sum_{n=1}^K (t_n)^2 - \left(\sum_{n=1}^K t_n \right)^2 \right]}} \quad (21)$$

degradation process, and changes in the operational state of the instrument due to environmental effects.³⁰ Therefore, a better performing HI is more robust to disturbances.

where t_n is the current monitoring time. From equations (20) and (21), the evaluation metrics that measure the robustness and the trend of the HI take values in the range $[0, 1]$, which have a positive correlation with the performance of the HI construction.

Step 3: Build the LSTM bearing prediction model. The training set and test set samples of the bearing RUL prediction model are divided using the bearing health indicators from step 2 as the input to the model.

Step 4: Optimize the hyperparameters in the CNN-LSTM network using the GTO algorithm to obtain the optimal prediction model. The optimization flowchart is shown in Figure 4 and is described as follows:

- (1) Initialize the relevant parameters of the GTO and CNN-LSTM networks. These include the number of gorilla populations, the maximum number of iterations, the convolutional kernel size and filters of convolutional layer 1, the size of pooling layer 1, the convolutional kernel size and filters of convolutional layer 2, the size of pooling layer 2, and hyperparameters such as Batchsize, the number of hidden layer neurons, and the learning rate in the LSTM network.
- (2) Based on the average square difference between the prediction value and the actual value of the neural network, a set of hyperparameters can be found to minimize the prediction error.
- (3) Calculate the initial fitness value: based on the mean square difference size, update the gorilla's position according to equations (8), (14) and (16), calculate the fitness value of the new position of the individual, and compare the updated fitness value with the original optimal value to achieve the update of the population with the global optimal solution.
- (4) Step 2 is repeated until the iteration count reaches the termination condition, outputs the optimal parameters, and obtains the optimal CNN-LSTM model.
- (5) Predict the bearing RUL using the optimal model obtained in step 4 and verify the accuracy of the model.

Experimental demonstration

Introduction of the experimental platform

Experimental data on bearings is obtained from the joint health monitoring laboratory for mechanical equipment set up by the Institute of Design Science and Basic Components at Xi'an Jiaotong University, and Changxing Sumyoung Technology Co., Ltd.³³ The structure of this test platform can be seen in Figure 5, which consists primarily of an alternating current motor, a motor velocity controller, a sustain shaft, two sustainer bearings, and a hydraulic load system. Two PCB 352C33 accelerometers are placed on the housing of the bearings, both accelerometers are placed at 90° to each other, that is, one on the vertical axis and the other on the horizontal axis. A DT9837 portable dynamic signal collector was used to collect vibration

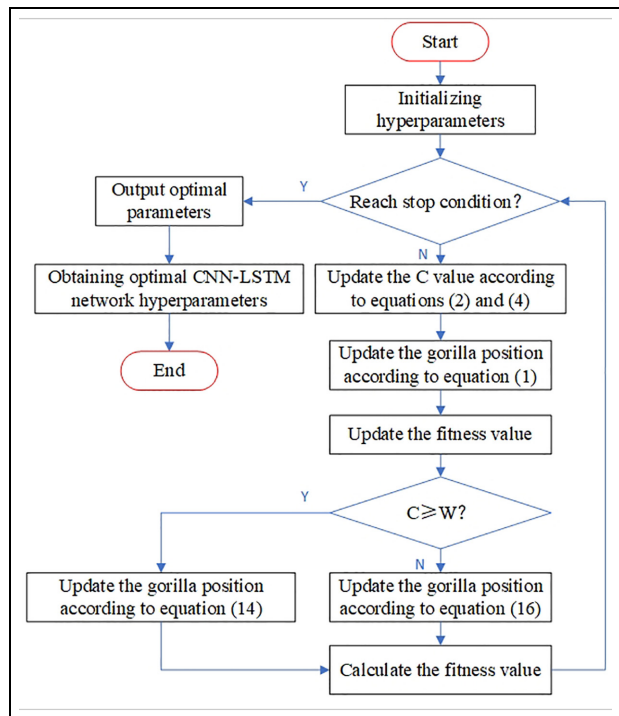


Figure 4. GTO algorithm optimization CNN-LSTM flow chart.

signals during the test. The sampling frequency was 25.6 kHz, the sampling interval was 1 min, and the duration of each sample was 1.28 s. The sampling frequency was 25.6 kHz.

Figure 5 illustrates the test bed used for the dataset and this test platform is typically used for accelerated bearing degradation testing.

The original vibration signal is collected by an acceleration sensor mounted on the outer ring of the test bearing, and a radial load is applied to accelerate the degradation of the bearing. This data set contains primarily 15 vibration test data sets for the entire bearing life cycle. Specific information on operating conditions and bearing life is given in Table 1.

Because the load is applied in the horizontal direction, the vibration signal in this direction can contain more degradation information, so the horizontal vibration signal is selected for the main analysis.³⁴

Figure 6 shows the horizontal vibration signal of Bearing 1_1. From Figure 6, the bearings oscillation signal amplitude changes significantly with time, and the signal turns into the scattering trend, which helps to extract the deterioration trend health indices.

The effect of different hyperparameters on the extraction of features

Twenty time and frequency domain features are selected as the input of CNN to extract the deep features hidden in the vibration signal, in order to fully utilize the feature information in the vibration signal. And the 20 features are reduced to one-dimensional

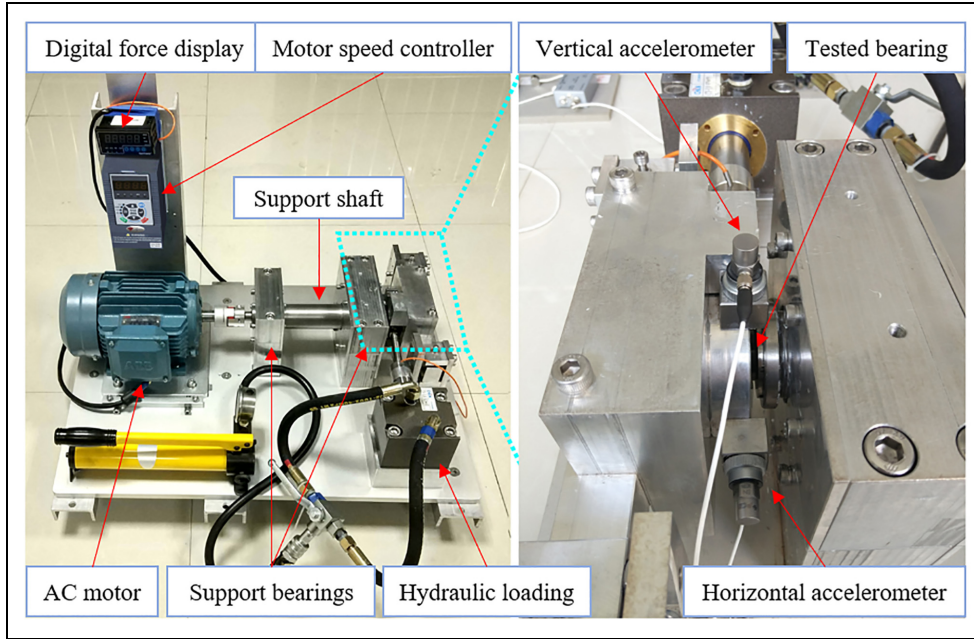


Figure 5. Bearing experimental platform.

Table 1. The specific information of operation conditions and bearing life.

Operation condition	Bearing dataset	Bearing life (min)
Condition (1) (12 kN/2100 rpm)	Bearing1_1	123
	Bearing1_2	161
	Bearing1_3	158
	Bearing1_4	122
	Bearing1_5	52
Condition (2) (11 kN/2250 rpm)	Bearing2_1	491
	Bearing2_2	161
	Bearing2_3	533
	Bearing2_4	42
	Bearing2_5	339
Condition (3) (10 kN/2400 rpm)	Bearing3_1	2538
	Bearing3_2	2496
	Bearing3_3	371
	Bearing3_4	1515
	Bearing3_5	115

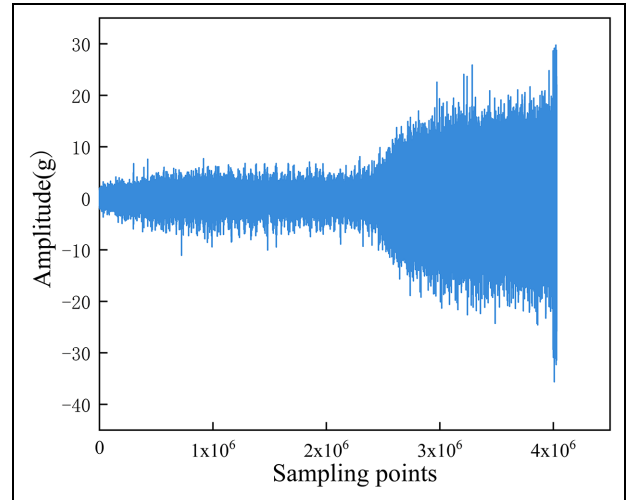


Figure 6. Horizontal vibration signal of the bearing.

features to obtain real-time data features and failure features. These include 16 time-domain features such as root mean square, root mean square, max, min, peak-to-peak, waveform coefficient, and more. Whereas 13 features in the frequency domain primarily include center frequency, root mean square frequency variance, and so on. Taking Bearing 1_1 as an example, a two-layer CNN network is constructed, and the CNN network hyperparameters are empirically set as in Table 2.

Figure 7 first shows the real-time features for different numbers of convolutional kernel channels.

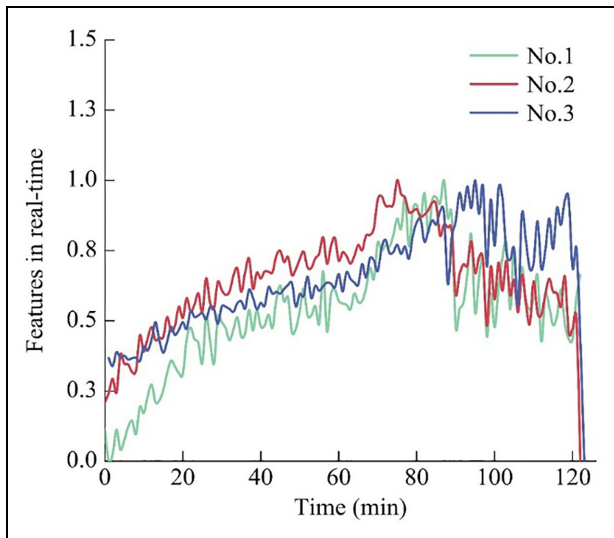
As can be seen in Figure 7, the overall trend of real-time features extracted by the CNN network is approximately the same for the same convolutional kernel size. When the filters of convolution layer 1 is 8, the

first change with large fluctuations in real-time characteristics occurs at 75 min. Comparing the bearing horizontal vibration signal is found to correspond to the moment when the vibration signal amplitude becomes significantly larger for the first time, indicating that it is more sensitive to early bearing failure. However, when the filters of convolution layer 1 increase, the first change with large feature fluctuations occurs at 88 min, indicating that it is more sensitive to the mid-term bearing failure. In addition, when the filters of convolution layer 2 are increased, the initial value of real-time features increases, and the feature value returns to 0 at the last moment.

Figure 8 shows the effect of the change in convolutional kernel size on the real-time features for the same

Table 2. CNN network hyperparameters.

No.	Convolutional layer 1		Convolutional layer 2	
	Kernel size	Filters	Kernel size	Filters
1	$[3 \times 1]$	8	$[3 \times 1]$	16
2	$[3 \times 1]$	8	$[3 \times 1]$	32
3	$[3 \times 1]$	16	$[3 \times 1]$	32
4	$[3 \times 1]$	8	$[3 \times 1]$	16
5	$[3 \times 1]$	8	$[5 \times 1]$	16
6	$[7 \times 1]$	8	$[5 \times 1]$	16
7	$[3 \times 1]$	8	$[5 \times 1]$	16
8	$[11 \times 1]$	4	$[9 \times 1]$	8

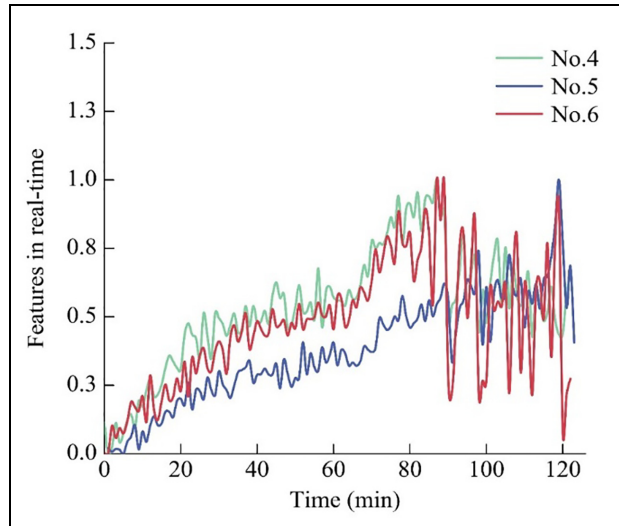
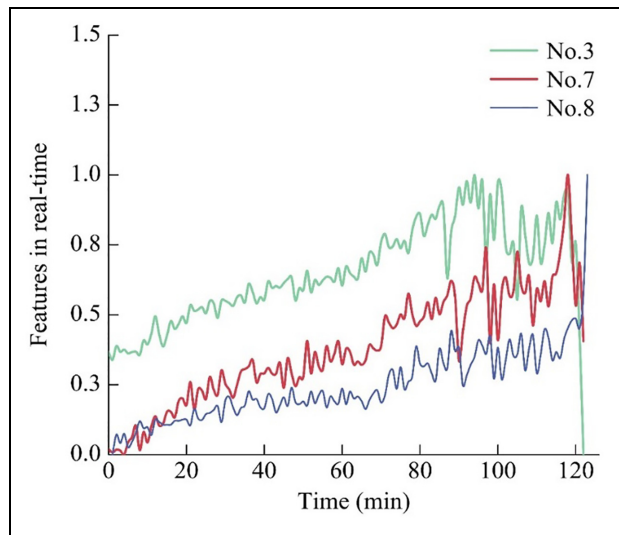
**Figure 7.** Real-time characteristics with different filters.

filters of the convolutional kernel (8 filters in the first layer and 16 filters in the second layer).

Figure 8 illustrates that the general trends of the real-time characteristics are the same for the same convolution kernel. However, when the convolutional kernel size of the convolutional second layer is increased, the real-time features show a roughly monotonic trend, peaking at around 119 min, which is generally consistent with the pattern of the real-time features. It indicates that as the size of the convolution kernel in the second layer increases, it may be beneficial for feature extraction. When the size of the convolutional kernel of convolutional layer 1 increases, the real-time features show a monotonic trend until 88 min but fluctuate too much in the later period.

Figure 9 points out the performance of the real-time features when the filters of the convolutional kernel and size change simultaneously.

The trend of the first half of the real-time features in Figure 9 is approximately the same, but the starting point is different, and in addition, the moment of the peak in the second half is different from the last moment of the feature data. In summary, when the

**Figure 8.** Real-time features with different convolution kernel sizes.**Figure 9.** Real-time features with different filters and sizes of convolutional kernels.

number and size of convolutional kernel channels in the convolutional layer of the CNN network change, the extracted real-time features also change. Based on the many hyperparameters of CNN-LSTM networks and the difficulty of selecting them, this paper proposes an adaptive method for selecting the filters and size of convolutional kernels in combination with GTO optimization algorithm and applies it to the remaining life prediction of bearings.

Construction of health indicators

The optimized CNN network hyperparameters obtained by combining the RUL prediction results are shown in Table 3. The health metrics are constructed

Table 3. GTO-CNN hyperparameter setting.

CNN network layer	Hyperparameter setting
Convolutional layer 1	Kernel size = [3, 1], filters = 16
Pooling layer 1	6
Convolutional layer 2	Kernel size = [7, 1], filters = 32
Pooling layer 2	12

based on the real-time features extracted in 4.2 using equation (19). The health indicators are shown in Figure 10.

Figure 10 shows the health indicators of different bearings under three operating conditions, with 1 representing the bearing in a healthy state and 0 representing the bearing in a failed state.

To verify the advantages of the proposed health indicators, traditional feature dimensions reduction methods Principal Component Analysis (PCA) and CNN networks based on empirical settings were used as comparative methods for determining health indicators, respectively. The initialization parameters of conventional CNN networks are set according to the literature.³⁵ The closer the robustness and trend are to 1, the better the bearing health index is. The results of robustness and trend of each bearing health index calculated

using this paper and the two comparative methods are shown in Table 4. The results show that the robustness and health index of the proposed method is greater than the other two comparative methods, which is beneficial to RUL prediction.

RUL prediction and comparative study

After constructing the health indicators based on the GTO-CNN network, the optimized LSTM network is input for lifetime prediction. Where the network hyperparameters are set as shown in Table 5.

Take Bearing 1_3 for example, health indicator length 158. Fifteen consecutive data were taken as a time series with a moving step of 1.³² The latter 30 min were taken as the test set, and the HI estimation and prediction results are shown in Figure 11. The blue dashed line indicates the HI prediction value, and the predicted RUL distribution is fitted using Monte Carlo simulation technology to give the 95% confidence interval (CI) of HI prediction and the probability density function (PDF) of RUL, which effectively solves the problem of uncertainty of prediction results in common deep learning models that are difficult to measure and provides a reliable basis for subsequent health management.

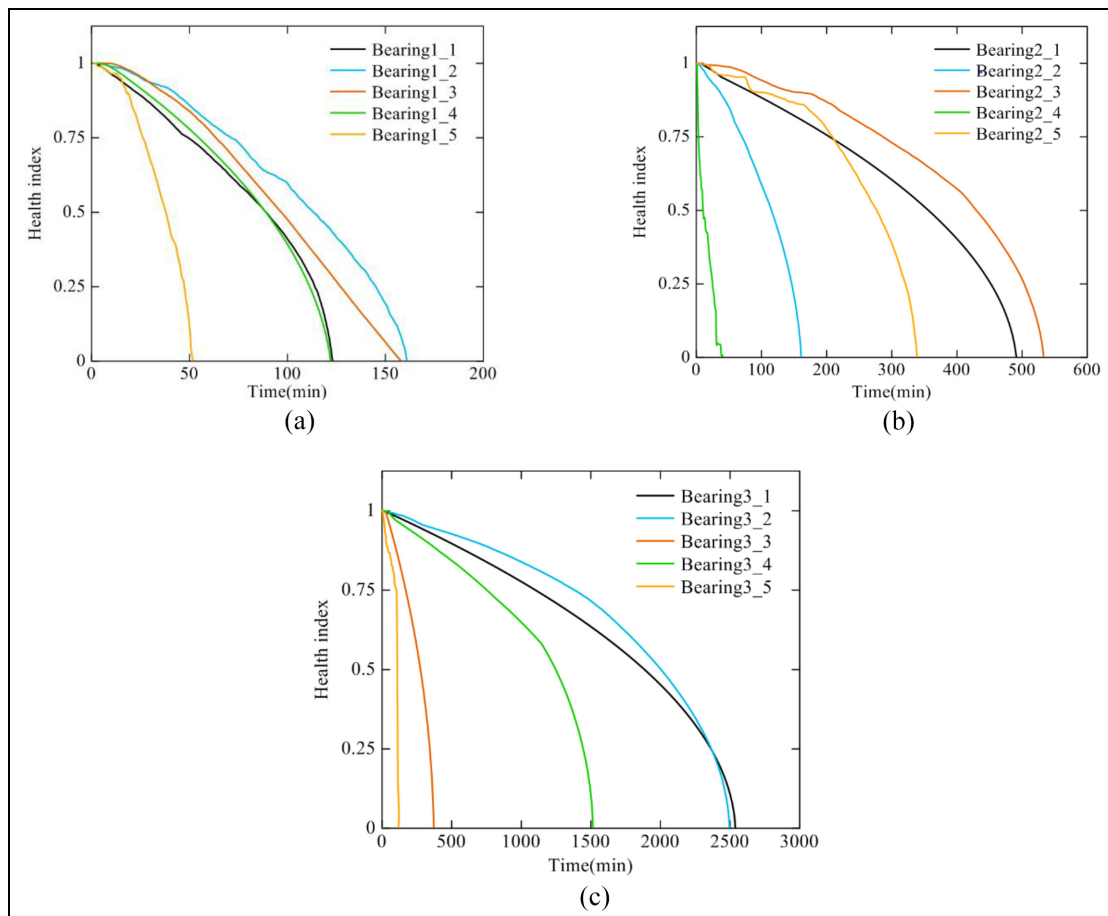
**Figure 10.** Health indicators under three working conditions: (a) Condition 1, (b) Condition 2, and (c) Condition 3.

Table 4. The performance measure comparisons by three methods.

Testing bearing	GTO-CNN-HI		CNN-HI		PCA	
	Rob	Corr	Rob	Corr	Rob	Corr
1_3	0.9884	0.9911	0.9463	0.8910	0.8325	0.9394
1_4	0.9823	0.9778	0.9603	0.9293	0.7324	0.5581
1_5	0.9506	0.9703	0.9358	0.9118	0.7296	0.8448
2_3	0.9956	0.9173	0.9498	0.8352	0.9028	0.7569
2_4	0.9203	0.9803	0.8691	0.9789	0.6932	0.9396
2_5	0.9926	0.9271	0.9500	0.8336	0.8643	0.9062
3_3	0.9942	0.9777	0.9238	0.8773	0.9589	0.5762
3_4	0.9985	0.9588	0.8026	0.6135	0.8735	0.2683
3_5	0.9696	0.8020	0.9407	0.7827	0.7885	0.6991

Table 5. Optimized LSTM network hyperparameters.

LSTM network layer	Hyperparameter setting
Batchsize	34
Hidden layer 1	59
Dropout	0.2
Hidden layer 2	70
Dropout	0.2
Learning rate	0.0063

Figure 11 shows that the predicted results for Bearing 1_3 HI reach the threshold 0 at 128 min, so their predicted RUL is 28 min, and the upper and lower intervals for reaching the threshold 0 at 95% CI are [124, 143] and the predicted RUL interval is [6, 13]. The results of predicting the RULs for Bearing 2_5 show the universality of the method proposed in this paper under different working conditions, with the confidence intervals narrowing as the sample data grows, indicating a closer approximation to the actual remaining service life.

Figures 11 and 12 show the RUL prediction results and 95% confidence intervals for Bearings 1_3 and Bearings 2_5, respectively.

The GTO-CNN-LSTM consists primarily of two parts, namely the enhanced CNN and the LSTM network. To reflect the performance prediction benefits of the optimized LSTM network, we compare this network to LSTM and CNN-LSTM networks commonly used for time series prediction, respectively, and show the resulting prediction result errors in Table 6. Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R2) are used as the model evaluation indexes. RMSE is used to reflect the deviation between the real value and the predicted value, and MAE represents the average of the absolute value of the error between the predicted value and the real value. The smaller the values of RMSE and MAE are, the lower the prediction error is in the model. R2 is used to assess the degree of conformity of

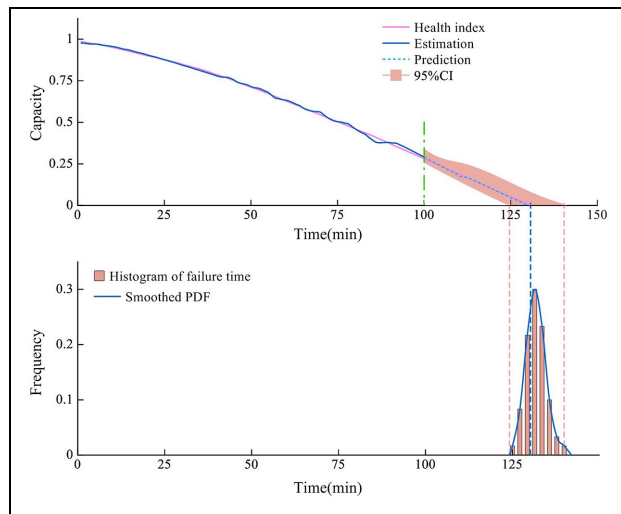


Figure 11. Bearing 1_3 RUL prediction results.

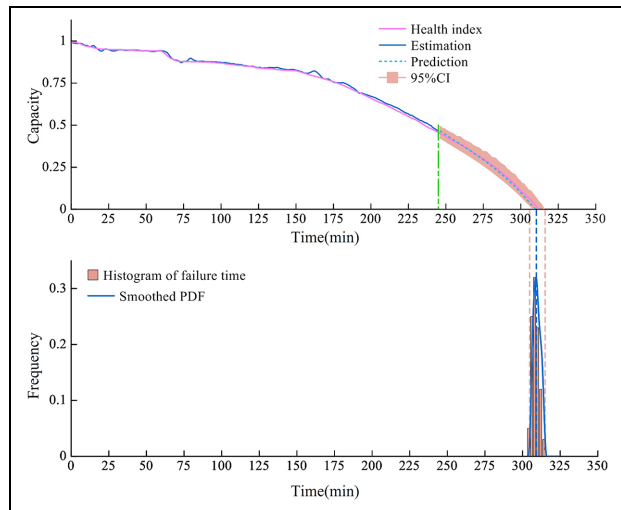


Figure 12. Bearing 2_5 RUL prediction results.

predicted value with the real value, and the larger the value is, the better the fitting effect is.

Table 6. The performance measure comparisons by three methods.

Testing bearing	LSTM			CNN-LSTM			GTO-CNN-LSTM		
	RMSE	MAE	R2	RMSE	MAE	R2	RMSE	MAE	R2
1_3	0.176	0.151	0.242	0.061	0.049	0.646	0.032	0.018	0.919
1_4	0.199	0.158	0.284	0.049	0.033	0.863	0.014	0.012	0.992
1_5	0.307	0.270	0.242	0.224	0.164	−1.854	0.034	0.029	0.945
2_3	0.336	0.230	0.460	0.077	0.051	0.819	0.023	0.011	0.988
2_4	0.193	0.192	0.871	0.152	0.125	−0.631	0.043	0.031	0.796
2_5	0.414	0.312	0.866	0.045	0.037	0.923	0.030	0.023	0.976
3_3	0.181	0.137	0.523	0.101	0.086	0.451	0.024	0.020	0.984
3_4	0.215	0.159	0.086	0.076	0.062	0.746	0.012	0.009	0.996
3_5	0.303	0.201	0.337	0.169	0.087	0.311	0.085	0.073	0.885
Mean	0.258	0.201	0.435	0.106	0.077	0.253	0.033	0.025	0.942

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (RUL - \hat{RUL})^2} \quad (22)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |RUL - \hat{RUL}| \quad (23)$$

$$R2 = \frac{\sum_{i=1}^N (\hat{RUL} - RUL)^2}{\sum_{i=1}^N (RUL - \bar{RUL})^2} \quad (24)$$

where RUL and $\hat{RUL}$ mean the actual RUL and the estimated RUL for the i th testing dataset, and the $\bar{RUL}$ is the mean of the actual RUL.

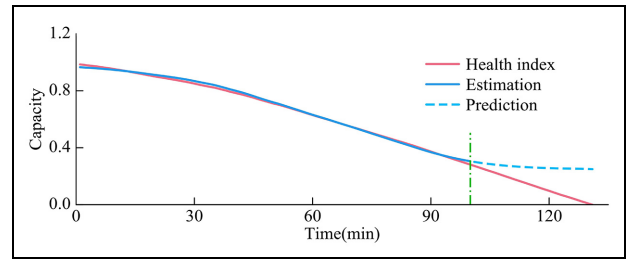
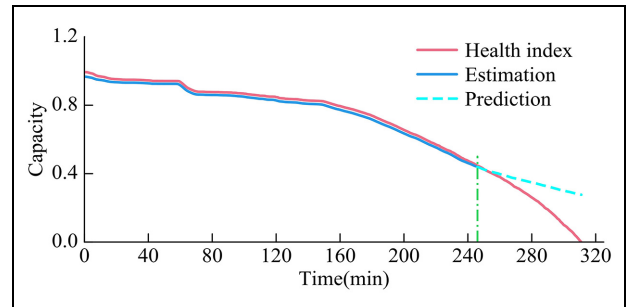
From Table 6 shows that the proposed method had a lower error ratio, demonstrating the superior performance of the model.

On the basis of obtaining health metrics, Bayesian optimization algorithm is used to optimize the hyperparameters in the CNN-LSTM network, such as learning rate, hidden layer nodes, and so on. Taking the test set Bearing1_3 and Bearing2_5 as an example, the optimal parameters are determined after 30 tests, and the following figure shows the prediction results.

From the Figures 13 and 14, it can be seen that the optimization of CNN-LSTM prediction results using Bayesian optimization algorithm (Bayes-CNN-LSTM) is not satisfactory. Table 7 demonstrates the results of the quantitative analysis of the two methods and the results show that the method proposed in this paper is more accurate.

Conclusion

To address the core problem of numerous and difficult to select network hyperparameters in RUL prediction, a GTO algorithm is proposed to optimize the adaptive prediction method of the CNN-LSTM network to predict RUL using the massive information in the whole

**Figure 13.** Bearing1_3 RUL prediction results.**Figure 14.** Bearing2_5 RUL prediction results.

life cycle of bearings, which improves the accuracy of RUL prediction, and validated by the Xi'an Jiaotong University bearing public dataset.

- (1) In the feature extraction process, the number and size of the convolutional kernel channels of CNN networks change, and the obtained real-time features also change, and the pattern is difficult to explain. In this regard, a feature extraction method with an adaptive selection of the number and size of convolutional kernel channels is proposed. Based on the GTO-CNN-LSTM network to analyze the monitored performance degradation data, the HI reflecting the deeper features implied by the degraded devices is extracted with the objective of minimizing the mean squared

Table 7. The performance measure comparisons by GTO-CNN-LSTM and Bayes-CNN-LSTM.

Testing bearings	GTO-CNN-LSTM		Bayes-CNN-LSTM	
	RMSE	MAE	RMSE	MAE
Bearing1_3	0.01892	0.00933	0.14966	0.13104
Bearing1_4	0.01624	0.01283	0.16492	0.13459
Bearing1_5	0.03929	0.02747	0.16535	0.12430
Bearing2_3	0.01421	0.00692	0.23252	0.16648
Bearing2_4	0.09686	0.06162	0.09321	0.08749
Bearing2_5	0.01410	0.00783	0.12206	0.09146
Bearing3_3	0.01536	0.01079	0.18758	0.14186
Bearing3_4	0.00846	0.00355	0.22807	0.17754
Bearing3_5	0.05373	0.04035	0.25929	0.14553

deviation of RUL prediction, and the HI extracted by the method in this paper has better robustness and trending compared with the commonly used dimensionality reduction methods.

- (2) In the RUL prediction stage, the optimized LSTM network is used to construct the time series information of HI, and the interval estimation of RUL is obtained through MC simulation technology, which can get more accurate RUL prediction results and distribution forms, and the results can be used for the subsequent operation planning, maintenance decision and other health management activities of bearings. In the RUL prediction stage, this paper used the optimized LSTM network to mine the time series information of the constructed HI and obtained the interval estimation of RUL by MC simulation technique, which can obtain more accurate RUL prediction results and distribution forms, and the results can be used for the subsequent operation planning, maintenance decision, and other health management activities of the bearings. In future research, a new network model will be considered to extract multiple health indicators to react to the health status of the device through different unsupervised learning models to further utilize the obtained monitoring data effectively, in addition, the Wiener process is considered to be introduced into the prediction instead of the LSTM model to achieve the purpose of number-model linkage, thus improving the accuracy of the remaining life prediction results.

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